StereoRetro Master Dataset — Data Card

`master_v1` — Build Record and Quality Report

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Abstract

This data card documents the construction, validation, and known limitations of `master_v1`, the frozen reaction-fact table underlying StereoRetro. It reports every threshold, every measured quality rate, and every documented gap from the build process, following the principle that a dataset is defensible not because it is free of error but because its errors are measured, bounded, and disclosed.

Contents

1 Source Composition

Four independent reaction sources were processed through one shared core pipeline (recombine → standardize → strip counterions → atom-map with RXNMapper → derive facts) and merged once.

Source	Raw reactions processed	Native role information
ORD (`dataset_phd_final.csv`)	2,375,686	`role_source` (reactant/catalyst/solvent blocks)
USPTO-50K	50,016	Weak positional (`reactants>reagents>product`)
USPTO-MIT	479,020	None (single `reactants»product` pool)
USPTO-STEREO	1,002,967	Weak positional (`reactants>reagents>products`)
Total, pre-merge	**3,907,689**	

All four sources were atom-mapped with the **same** RXNMapper instance (never the sources' own bundled mappings, e.g. USPTO-50K/MIT's Indigo-derived maps, which were stripped and discarded before re-mapping) – ensuring a single, consistent definition of "reactant" across the entire master.

2 Atom Mapping Quality

Metric	**ORD**	**50K**	**MIT**	**STEREO**
Fragment order preserved	26.6%	66.5%	25.7%	27.3%
Matched by canonical SMILES	97.0%	100.0%	99.0%	97.1%
Low-confidence ($\tau < 0.5$)	38.2%	29.8%	33.4%	40.0%
Element-mismatch (mapper error)	0.0%	0.0%	0.0%	0.0%

Fragment order preserved is the fraction of reactions where RXNMapper's output fragment order matched the input order. It is reported precisely because it is *low* across three of four sources (25.7–27.3%): this proves that any pipeline aligning roles to mapping output *by list position* would be wrong on the majority of reactions. All role/atom-contribution facts in this master are matched by **canonical SMILES**, never by index – this was found to be necessary, not a precaution, after an earlier index-based version produced materially different (and wrong) validation numbers (71% vs. 97% agreement on the same sample) before the fix.

Element-mismatch (an atom mapped to a product atom of a different chemical element – physically impossible, e.g. a Pd atom "contributing" to an organic product) was **0.0% on all four sources, at full scale**. This is the strongest available evidence that RXNMapper is chemically self-consistent on this corpus; it was used throughout the build as a sanity signal, not merely a diagnostic side-note.

3 Role Separation — Validation Against Existing Labels

For ORD, which ships its own role information (`role_source`: reactant/catalyst/solvent blocks), the atom-contribution fact (`atoms_contributed == 0` ⇒ spectator) was cross-checked against those existing labels.

Validation	**All reactions**	**High-confidence only**
Role label vs. atoms_contributed==0 agreement	93.7%	96.9%
Spectators hidden in raw `reactants` column	40.9%	42.0%

Interpretation. 96.9% agreement on high-confidence mappings means ORD's original solvent/catalyst labelling was largely correct – most disagreement traced, on manual inspection, to atom-index misalignment in an earlier script version, not to genuine labelling error (see §2). The remaining disagreement is dominated by chemically ambiguous molecules whose role is contextual by nature (acetic acid, water, methanol – sometimes solvent, sometimes reactant), consistent with the field's known reactant/reagent fuzziness, not a defect in either method.

The 40–42% figure is the single most consequential number in this data card. It means that roughly two in five molecules sitting in the raw `reactants` column of the source data are, by atomic contribution, actual spectators (bases, salts, catalysts). Any task view that uses the raw `reactants` column as a training target without deriving roles from `atoms_contributed` would teach a retrosynthesis model to generate spectators as if they were reactants.

4 Deduplication

Deduplication was performed on the canonical reaction key (`rxn`), **after** fact computation and **before** any split assignment, in two passes as sources became available.

Pass	Rows before	Rows after
Pass 1 (ORD + USPTO-50K + USPTO-MIT)	2,904,722	1,823,789
Pass 2 (+ USPTO-STEREO)	2,826,756	2,523,137

Duplicate origin was decomposed, not assumed. In Pass 1, of 1,628,424 rows sitting in duplicate groups, **459,873 groups were single-source** (predominantly ORD's own internal near-duplicate patent-optimization series, consistent with the ∼34% internal duplication already flagged during ORD's original curation) versus **87,618 groups showing genuine cross-source overlap**. In Pass 2, the pattern inverted: only 548 single-source duplicate groups remained (the three earlier sources were already internally clean) against **302,829 cross-source duplicate groups** – almost entirely USPTO-STEREO overlapping with the three prior sources, consistent with STEREO's own patent-derived provenance. **Net new reactions contributed by USPTO-STEREO: 699,348** (out of 1,002,967 raw, i.e. ∼70% of STEREO was genuinely new to the master).

5 Reaction Center

Computed post-hoc from `mapped_rxn` (no new mapping call): an atom is in the reaction center if its bonding environment (heavy-atom neighbors, bond types, **and hydrogen count**) differs between reactants and products.

Metric	3-source pass	4-source total (STEREO)
Reaction center computed	97.3%	97.7%
Empty reaction center (no bond change detected)	1.0% (18,174)	0.6% (5,841)
Median / mean center size (atoms)	2 / 2.4	2 / 2.8

A hydrogen-count blind spot was found and fixed during construction. The initial version compared only heavy-atom neighbors and bond types, which is blind to deprotection/protonation events that preserve the same heavy-atom connectivity while changing H-count (e.g. a Cbz-protected amine, N-H → N-H_2 upon deprotection). This produced 310,767 empty reaction centers (17% of the 3-source corpus) on the first run. Manual inspection of a sample confirmed the mechanism (verified against a real corpus row, an amino-acid Cbz deprotection); the fix (adding `GetTotalNumHs()` to the environment comparison) reduced empty reaction centers to 18,174 (1.0%) on the same data – a 94% reduction, confirming the diagnosis rather than merely suppressing the symptom.

The residual 1.0–0.6% empty reaction centers are not an error. Manual inspection of a sample found these to be predominantly (a) **chiral resolutions** (an unassigned stereocenter becomes defined with no connectivity change – exactly the substrate for `stereo_change`'s `resolved` category, §6) and (b) **salt formation/dissociation** (e.g. a fumarate salt pairing, where the recorded "product" is a non-covalent adduct or its dissociated free base) – both legitimate cases outside the scope of a bond-based reaction center, not computation failures.

6 Stereo Change

Computed alongside reaction center: for every mapped stereocenter (by atom map number), classifies its fate across the reaction arrow.

Category	3-source pass	4-source total (STEREO)
retained	1,024,973	533,162
created	141,737	81,224
destroyed	48,807	33,072
inverted	21,538	18,012
resolved	16,005	12,938
unresolved	14,220	10,217
Reactions with $\geq$1 stereocenter (either side)	702,969	383,667
Reactions with $\geq$1 status change (not just retained)	187,684	119,307

`retained` dominates in both passes, as expected: the majority of stereocenters survive a reaction unchanged. `created` exceeding `destroyed` by roughly 3:1 is consistent with typical synthetic chemistry (more reactions introduce a stereocenter, e.g. additions, than remove one, e.g. planarizing oxidations). **38.5% of the 3-source corpus (702,969/1,823,789) contains at least one stereocenter** – a substantial, directly usable subset for stereochemistry-focused evaluation.

7 Chiral-Catalysis and Stereo-Relevant Spectator Facts

Two independent, non-merged facts about spectator molecules (`atoms_contributed == 0`), addressing the known failure mode where a chiral catalyst/auxiliary contributes zero atoms yet causes the product's stereochemistry.

Fact	3-source pass	4-source total (STEREO)
Reactions with $\geq$1 stereocenter-bearing spectator	57,653	40,603
`reaction_chiral_catalysis_signal` = True	18,533 (1.04%)	16,526 (1.69%)

Metal list used for `reaction_chiral_catalysis_signal`: Pd, Rh, Ru, Ir, Cu, Ni. Not an arbitrary choice: this is the convergent core list across multiple independent literature sources consulted before implementation (an academic review of Pd-catalysed asymmetric allylic alkylation; a 2022 review on synergistic dual transition-metal asymmetric catalysis explicitly tracing the paradigm to the 2001 Nobel Prize in asymmetric catalysis). Broader lists in the same sources also mention Fe, Ti, Pt, Mo, W as usable but less consensus metals – deliberately excluded to keep the flag conservative rather than over-inclusive. The ∼1–1.7% flagging rate is consistent with asymmetric catalysis being a specialized minority technique within a general-purpose patent/reaction corpus, not the norm; STEREO's marginally higher rate (1.69% vs. 1.04%) is consistent with its name and intended composition.

8 Scaffold and Timestamp (Split Inputs)

Per the master's fact-vs-choice principle, no split (train/val/test) is assigned in the master itself. Only the raw inputs a split function would need are stored.

Metric	Value
Scaffold computed (Bemis–Murcko of product)	100.0% (2,523,137/2,523,137)
Acyclic products (valid empty scaffold)	3.9% (97,458)
Unparseable products (scaffold = None)	0.0% (0)
Unique scaffolds	273,974
Mean reactions per scaffold	∼9.2

Known gap: `timestamp` is `None` for every row. None of the four sources, as currently extracted, carry a native publication/patent date. This is not fabricated or approximated – it is left explicitly absent and disclosed here. **Consequence: scaffold-based splitting is available now; temporal/OOD splitting (as specified for the CSD-Retro-Stereo benchmark, 2022–2026) is not,** until `timestamp` is backfilled from source metadata (e.g. USPTO patent grant dates from patent IDs, or ORD provenance records) – a planned, additive gap-fill (one new column derived from data already identified), not a rebuild.

9 Final Master Composition

Quantity	Value
Total unique reactions (`master_v1`)	2,523,137
Sources represented	ORD, USPTO-50K, USPTO-MIT, USPTO-STEREO
Facts per reaction	mapping, roles, stereo, quality, reaction center, stereo change, chiral-catalysis signal, scaffold
Split assigned in master	None (by design – see §7)

10 Known Limitations (Disclosed, Not Hidden)

1. **No oracle exists for role assignment.** All validation is agreement between independent methods (atom-contribution vs. source-provided labels), never certified ground truth.

2. **RXNMapper's own published reliability ceiling (∼84% on independent benchmarks) bounds every fact derived from** `mapped_rxn`, including `atoms_contributed`, `reaction_center`, and `stereo_change`.

3. **H_2 and other zero-heavy-atom reactants** are structurally invisible to atom-count-based role assignment (no heavy atoms to map), a known limitation of the atom-contribution definition itself, not this implementation.

4. **No timestamp in any source** (§8) – temporal/OOD splitting is not currently possible.

5. **Sources are patent- and database-derived**, not exploratory academic chemistry – known compositional bias inherited from USPTO/ORD, not correctable by cleaning.

6. **The reactant/reagent boundary remains conventionally fuzzy** for molecules that contribute atoms but are chemically termed reagents (e.g. methylating agents) – this master resolves it by the atom-contribution definition throughout, documented as a stated convention, not a claim of resolving the ambiguity itself.

This dataset is not, and does not claim to be, error-free. Its defensibility rests on every threshold, disagreement rate, and gap being measured and disclosed above – consistent with the standard that a Q1-defensible dataset is one whose limitations are known and bounded, not one presumed perfect.